# Who's Aligned to Whom? How AI Agents Shape Experts' Research Judgment Calls

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**Abstract**

As AI systems become more capable at tasks with verifiable answers, experts' role in knowledge work is shifting to judgment calls, such as which analytical approach to take, how to interpret ambiguous evidence, and what options to recommend to decision-makers. If experts already defer to AI on these calls, perhaps without knowing it, AI priors will settle consequential decisions in science and policy, including in the research meant to keep AI safe and aligned. To test whether experts defer, we ran a randomized trial in which 50 causal-inference practitioners analyzed observational data with an AI coding agent and delivered a policy recommendation. On four analytical decisions and the final recommendation, a coin flip set which of two defensible options the participant's agent would lean toward. Participants did not know the agent was assigned a lean, and the agent made the best honest case for it. On the analytical decisions, participants went with the agent's lean 86% of the time (50% by chance), and, in an exploratory analysis, these leans alone predicted 82% of rollout-or-discontinue recommendations (majority baseline 61%). After the task, participants recalled most of their decisions accurately, and more than half rated the recommendation as more their own than the agent's. Yet when asked to justify their choices, 77% repeated the agent's talking points, sometimes nearly verbatim, and only 15% mentioned the agent in any answer. Whether they were persuaded or went along, participants let the agent's randomly assigned priors settle most of the analysis, and their own accounts often did not reflect this. If human judgment is to remain a meaningful check as more work is done with AI, we will need ways to measure deferral and to help experts recognize it.

## 1 Introduction

As AI systems become more capable at tasks with verifiable answers, such as writing code and running experiments, judgment calls are becoming a larger part of expert work. An expert generally still has to decide, for instance, which analytical approach to take, how to interpret ambiguous evidence, and what to recommend. Many of these judgment calls are underdetermined: the evidence supports more than one defensible answer, and the choice depends on the priors and assumptions of whoever makes the call. Experts may lean different ways on such calls, but so may the AI agents they work with, and we know little about how experts engage with agents' leans. We therefore ask to what extent experts defer consequential analytical decisions without ground truth to AI agents and whether that influence remains visible in their post-task accounts.

The literature on overreliance and AI-assisted decision-making asks how people can rely on automated decision systems just *the right amount*, and focuses on the cases where humans accept an AI recommendation that is wrong or override one that is right. The standard overreliance approach assumes the decisions have ground truth, so that overreliance, underreliance, and appropriate reliance can be defined (Vasconcelos et al., 2023). Appropriate reliance, in this tradition, is deference conditioned on

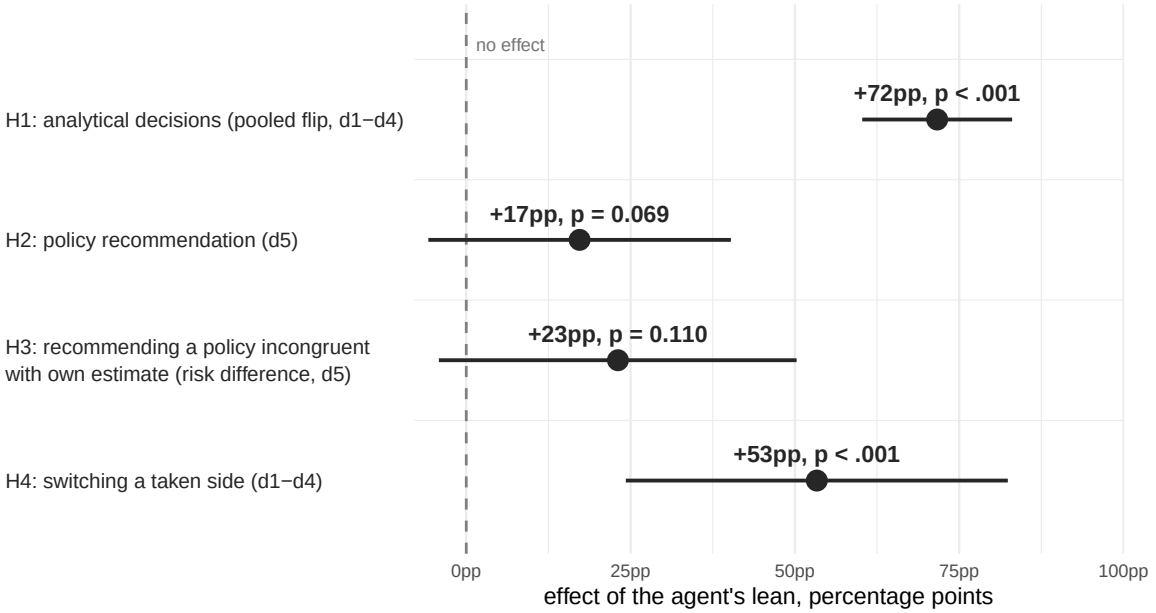



Figure 1: Estimates of the four registered quantities with two-sided 95% confidence intervals. The pooled flip rate (H1) is the difference between lean arms in the probability of choosing option A, averaged with equal weight over the analytical decisions $d_1$–$d_4$. The policy shift (H2) is the difference in mean recommendation score (discontinue 0, extension 0.5, rollout 1) between the rollout-lean and discontinue-lean arms on $d_5$. H3 compares departure rates (how often participants recommended against what their own estimate implied) under incongruent and congruent $d_5$ leans, among participants whose estimates were clearly above or below the 3–4% target. We report it descriptively because the agent's lean may have affected which participants reached such estimates, so the two lean arms within this group are no longer comparable; a registered check found evidence of this. The switch-rate gap (H4) compares opposed and aligned voiced leans in the qualifying subset of $d_1$–$d_4$ trials in which the participant took a side before the agent voiced its assigned lean.

correctness (Schemmer et al., 2023). The judgment calls above have no such ground truth, so overreliance cannot be defined for them. Instead, decision-making authority may drift from the expert to the agent, perhaps without the expert realizing it (Williams-Ceci et al., 2026), while the lack of immediate feedback makes it difficult for them to learn when the agent is reliable.

These situations resemble what researchers call *wicked learning environments* (Hogarth, 2001; Hogarth et al., 2015), where the operating rules are not clear and can shift, and feedback on actions is delayed, unreliable, or absent. The fluent output of AI models can look much the same whether the conceptual reasoning is sound or not (Steyvers et al., 2025). In human writing, words of certainty often serve social functions beyond expressing how certain the writer is, and models trained on that writing may learn to use them the same way (Zhou et al., 2023). Experts may keep using agents for the decisions where the agents' reliability and the right level of reliance are hardest to assess.

We use observational causal inference (OCI) as our testbed. OCI frequently requires practitioners to work with incomplete data that can support contrasting hypotheses and forces them to make judgment calls such as which assumptions to rely on, how to operationalize the primary outcome, and how to define inclusion and exclusion criteria (Silberzahn et al., 2018; Breznau et al., 2022; Huntington-Klein et al., 2025). These judgment calls give experts real freedom, but not all of this freedom is worth protecting.

While some of it is room for arbitrariness, even error, that an agent can usefully curb (Steegen et al., 2016; Simonsohn et al., 2020; Huntington-Klein et al., 2025), some of it is room for the judgment experts are relied on for.

If experts follow an agent's lean on these decisions, intentionally or not, the agent's priors become an unacknowledged determining factor in the analysis. A recommendation looks the same whether the expert reached it on their own or adopted it from the agent, so deferral is not obvious from the delivered work. We conducted a randomized controlled trial where professional practitioners worked on a realistic analysis task, built on synthetic data, with multiple underdetermined decision points. Unknown to the participants, the AI agents they used were randomly assigned to lean strongly toward one option and were instructed to express this lean. This design lets us estimate the causal effect of the agent's lean on the final decisions the participants made. Specifically, we computed the average net influence of an agent's lean on participants' decisions, which we call the *flip rate*. The flip rate is a measure of influence and does not require the decisions in the task to be labeled right or wrong.

### 1.1 Summary of our results

In our experiment, participants analyzed observational data from a fictional farming cooperative, estimated the productivity gains of a voluntary farm-advisory program, and, based on the estimate, provided a policy recommendation: roll out the program or discontinue it. The task involved two kinds of decisions that we tracked: four analytical decisions (such as operationalizing productivity and choosing the causal identification strategy) and one final policy recommendation, each with a randomly assigned agent lean.

Figure 1 and Table 2 report the registered estimates. On analytical decisions, an agent's lean can flip a participant's final choice 72% of the time. Put differently, participants' choices matched the agent's lean 86% of the time, against a chance rate of 50% under random assignment. On the subset of trials where the participant had visibly taken a side before the agent voiced its lean, an opposing voiced lean made switching away from that side 53 percentage points more likely than an aligned one. The shift on the policy recommendation was smaller and not statistically significant under the registered primary estimator: 17% of the distance from discontinuing to rolling out (22% unadjusted). The smaller shift does not mean participants retained control over the recommendation: in an exploratory, in-sample analysis, the four intermediate leans alone correctly predict 82% of delivered recommendations among participants who did not choose an extension (61% majority baseline; AUC 0.84), because the estimate the recommendation relies on is already shaped by the analytical decisions.

## 2 Background

The effects of automated decision systems on human behavior have been studied for decades under concerns such as oversight, overreliance, control, and automation bias. Historically, this work has presumed a correct choice, which lets overreliance, underreliance, and appropriate reliance be defined. Our work builds on the research strands around automation bias, AI overreliance, and AI-induced disempowerment. We ask what overreliance means when the decisions experts face are underdetermined and no choice is right or wrong. We first review the reliance tradition and its ground-truth assumption, then the effects of researcher freedom in underdetermined calls where that ground truth is missing, the conditions under which such calls pass to the agent unnoticed, and finally the costs of repeated deferral.

The modern overreliance literature mostly keeps the assumption of the existence of a correct answer. Its focus is whether people can detect wrong AI recommendations and override them (Schemmer et al., 2023). This literature studies trust in AI, evaluation skills, and interaction design as levers for finding the right amount of reliance (Lee and See, 2004; Lai et al., 2023). The assumption extends to adversarial

settings, where the agent itself may sabotage the work (Ye et al., 2026). In this context, expertise is a decisive determinant of the ability to detect and override a bad recommendation.

In recent years, multiverse and specification-curve analyses have shown how much researcher degrees of freedom can influence results (Steegen et al., 2016; Simonsohn et al., 2020), and independent teams in many-analysts studies produce widely varying estimates from the same data and question (Silberzahn et al., 2018; Breznau et al., 2022).[1] Part of this freedom can be dysfunctional, even pathological, as in p-hacking and the garden of forking paths (Simmons et al., 2011; Gelman and Loken, 2014). Fixing decisions upfront, registering pre-analysis plans, and replication studies are how the scientific community has tried to address this issue. However, even explicit and prescribed attempts to impose discipline with tighter specifications can fail to decrease variation because teams may fail to implement the specifications properly (Huntington-Klein et al., 2025).

A more functional form of freedom is creativity, novelty, and flexibility in the analysis, which are at risk of homogenization when the same AI agent is used by multiple experts. Authors writing with the same model produce more similar text (Padmakumar and He, 2024), and generative AI raises individual creativity while reducing the collective diversity of what is produced (Doshi and Hauser, 2024). The effect can have serious implications for the scientific method, because apparent consensus may be misleading if different researchers use models with similar priors.

When outcomes cannot validate choices, expertise and credibility rest on procedural conformity and reasoning, which can be difficult to distinguish from argumentative fluency. An AI agent can produce both procedural conformity and fluent reasoning at near-zero cost, giving the impression of having expertise. An agent that arrives with a prior and voices it can therefore settle exactly the choices the procedure leaves open.

The cognitive offloading perspective, complementary to our approach, asks what part of a task's cognitive work is delegated to the tool (Padmakumar et al., 2026). When the offloaded work includes underdetermined calls, the agent must supply the assumptions and priors that settle them, and the nature of the delegated work transforms from execution to judgment.

Preferences are also partly constructed in the process of deciding, in elicitation (Slovic, 1995) and in deliberation and voting (Elster, 1997). Sometimes people do not know what they want. A decision can stay underdetermined because of that, and determining it can require a process through which the person discovers their preferences. An agent that settles such a decision also settles it in place of that process.

The contrastive-explanation literature defines explanation as answering why this outcome rather than the alternative – the first level of contrast (Lipton, 1990; Miller, 2019). A second level of contrast, why a competent expert would prefer the alternative and on what assumptions and values, settles an underdetermined decision. The second level has close parallels in scalable oversight, where a consultant assigned one side pulls judges toward that side while debate, which adds the second side, reduces the pull (Michael et al., 2023; Khan et al., 2024), and in dialectical inquiry, which requires stating the assumptions under which the counter-plan is correct (Mason, 1969; Schweiger et al., 1986).

Navigating underdetermined calls well is itself a mark of expertise, which raises the question of how practitioners build it. In the Dreyfus model of skill acquisition, mastery progresses from explicit rule following toward situational judgment (Dreyfus and Dreyfus, 1980), and experts arrive at a problem with heuristics and pattern recognition acquired before the data are ever seen (Klein, 1998; Hutton and Klein, 1999). The detrimental effects of automation on these skills were noted as early as the 1960s and 1970s. Bainbridge (1983) focuses on the tension between automating the operator's work and still depending on

[1] Readers interested in critiques and re-interpretations of the data of Breznau et al. (2022), which at a meta level provide a further example of researcher degrees of freedom, can refer to Mathur et al. (2023), Engzell (2023), and Auspurg and Brüderl (2024).

the operator's skill to intervene when the automation fails – a skill "which later generations of operators cannot be expected to have". Shen and Tamkin (2026) find a related effect with AI assistance: software developers who used it to learn a new library ended up with weaker conceptual understanding.

AI-induced disempowerment provides an interpretive frame for the risk that the agent's prior settles the choices the procedure leaves open. Chooi et al. (2026) define it as "The erosion of humans' ability to meaningfully shape outcomes that affect their lives, where such erosion is caused or substantially mediated by AI systems." It can manifest at the individual level (Sharma et al., 2026) or the systemic level (Kulveit et al., 2025). At the individual level, Sharma et al. (2026) define situational disempowerment as holding inaccurate beliefs about reality, making value judgments that are inauthentic to one's own values, or acting in ways misaligned with those values. Our design limits reality-distortion potential. The agent is instructed to present accurate information and not to mislead or hide data, leaving its framing and how it puts results together as the residual concern. By contrast, the potential to distort judgments is high. Where the disempowerment literature centers on beliefs, personal values, and preferences, the judgments at issue here are analytical – the decisions, the framing, and the reasoning used to weigh them. The potential to distort action is also high because the agent may lead experts to take actions that they would not have taken in the counterfactual. The relevant risk in our setting is therefore the deferral to an agent of the analytical judgments and actions through which experts produce their work.

## 3 Methodology

### 3.1 Participants

We recruited professional causal-inference practitioners through a paid expert-annotator marketplace, targeting N = 50 after exclusions. Candidates passed a five-part vendor screen (Appendix A.1). We stopped fielding when the registered stopping criterion was met.[2] In total, 61 participants completed the task with a valid deliverable; 11 of them did so without using the AI agent and serve only as a no-AI comparison group, leaving 50 participants in the sample, of whom 48 completed the pre- and post-task surveys. The final sample spans data science, mathematics, economics, political science, and AI/ML practice. Among participants who completed the surveys, 65% held a master's or doctoral degree or had six or more years of experience in analysis roles, and 90% rated their causal-inference expertise as proficient or expert. Table 1 summarizes the included sample, and Table 4 in Appendix A.2 reports the full descriptives.

| | n (%) |
|---|---|
| Included participants | 50 |
| Causal-inference expertise (self-rated) | |
| Proficient or Expert | 43 (90%) |
| Basic | 4 (8%) |
| Novice | 1 (2%) |
| Education | |
| Master's or Doctorate | 24 (52%) |
| Bachelor's | 20 (43%) |
| High school or some college | 2 (4%) |

Table 1: Participant characteristics of the included sample. Expertise is self-rated; all participants passed the knowledge check in the vendor screen (Appendix A.1), and Table 4 reports their scores on our skill test.

[2]By the time we stopped the experiment, an additional participant had completed the task.

### 3.2 TASK AND THE DECISIONS

The task is a fictional scenario, built on data from a data-generating process we control, where the participant acts as the principal analyst for the Greenfields Farming Co-operative, whose board is deciding whether to roll out or discontinue a subsidized, voluntary farm-advisory program. The board asks for a recommendation and states a target of a 3–4% productivity gain. The in-task briefing intentionally leaves the productivity measure, the identification strategy, the estimand, and the weighting open to judgment. Both the data and the in-task briefing support more than one reasonable choice for these decisions, and the decisions are consequential. Based on individual choices, the estimated effect size can range from strong productivity loss to strong gains.

We track five binary decisions an analyst must make, explicitly or implicitly, to deliver a policy recommendation. By convention, side A pushes the estimate up or toward rollout, and side B pushes it down or toward discontinuation.

- $d_1$, productivity measure (A = yield per hectare, B = yield per operating cost).
- $d_2$, identification (A = difference-in-differences, B = levels-adjusted ANCOVA).
- $d_3$, estimand (A = effect on enrolled farms, the average treatment effect on the treated, ATT; B = effect on the full membership, the average treatment effect, ATE).
- $d_4$, weighting (A = per-farm, B = production-weighted).
- $d_5$, policy recommendation (A = roll out to everyone, B = discontinue).

Appendix A.5 details, for each decision, why it is underdetermined, the grounds each side can stand on in the task materials, and the two priors written for the agent.

The policy recommendation is special because at the time of decision, the participant already has a clear anchor on which to determine the soundness of this decision. The board asks for a binary policy recommendation, but either the participant or the agent may propose a one-year extension instead. We retain the extension as a middle outcome between discontinuing and rolling out.

### 3.3 THE ASSIGNED LEAN

The lean amplifies a behavior of the agent that is already observed in real-world deployments. AI agents may, and do, have dominant baseline preferences in these decisions and may speak up and provide their recommendation in neutral queries, which we established via simulation-based results, summarized in the Results section. Every participant had an agent with leans assigned as described below. A system prompt, hidden from the participant, presents each lean as the agent's prior and encourages the agent to state it when the decision arises or the participant asks.

Appendix A.3 quotes the lean prompt's shared preamble and describes the built-in escape hatch that routes both $d_5$ priors to a one-year extension when the estimate contradicts the assigned side.

The agent follows the participant's instructions, and participants may choose any option, including options outside those listed. The agent is pinned and identical for all sessions (Claude Code v2.1.220 running Claude Opus 4.8 at high effort).

Each participant's agent receives one of the 32 possible configurations of the five binary leans. Configurations are drawn from randomized blocks that contain every configuration exactly once, with consecutive pairs taking opposite sides on all five decisions, so the five factors stay balanced throughout fielding. A configuration is assigned when a participant first enters the task environment; every released position is consumed even if the task is incomplete or later judged invalid, and positions are never reused.

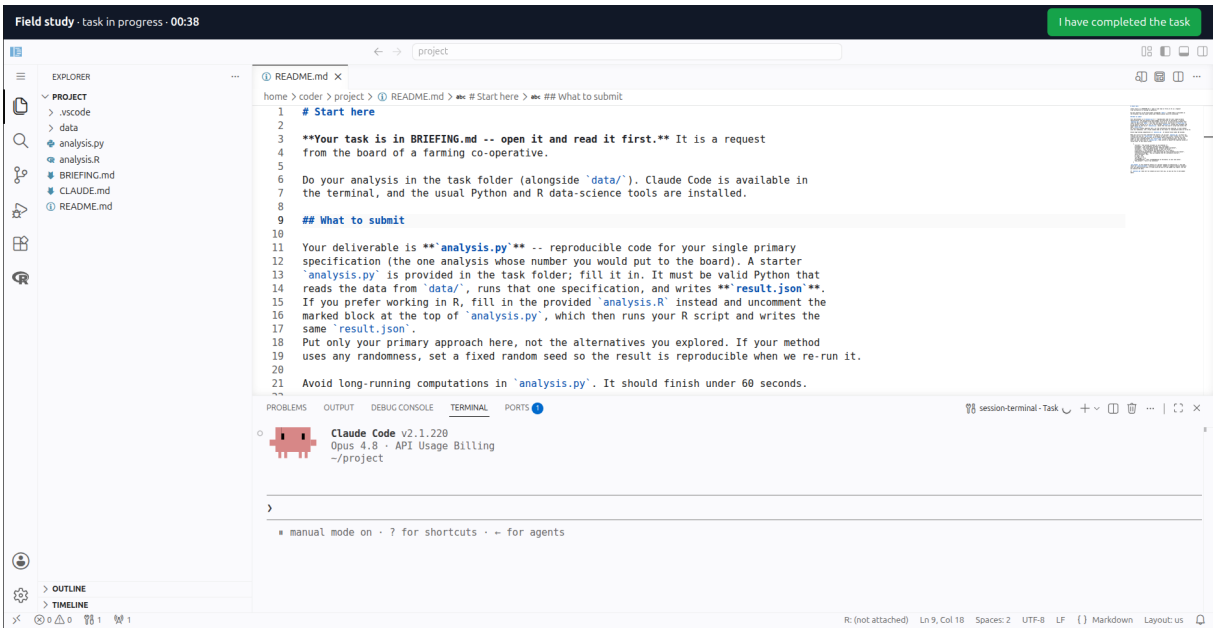



Figure 2: The task workspace: a browser-based VS Code environment with Claude Code available.

### 3.4 Procedure and measures

After completing the pre-task survey on the vendor's platform and waiting for a minimum 24-hour cooling period, participants entered the experimental platform through a unique URL. Upon task completion, they immediately returned to the vendor platform for the post-task survey. In the task, participants read the briefing and worked in a monitored browser workspace with a code editor, Claude Code, and the cooperative's observational records. They could use Python or R with a supplied boilerplate script. Figure 2 shows the task workspace. We elicited no decisions before agent use because elicitation is itself an intervention and can understate deference in naturalistic workflows (Buçinca et al., 2021; Fogliato et al., 2022). The platform recorded the transcript, screen, window focus, and inactivity. Participants were told to plan for roughly 60 minutes of uninterrupted work and that the task was self-paced. The task ended when participants submitted a structured JSON report containing their point estimate, confidence interval, and five choices.

The pre-task survey collects the fixed pre-task pool used by the registered covariate adjustment. The post-task survey opens with an open-ended question about the study's purpose, asked before any cueing, which we use for the registered awareness check, then collects recall, per-decision choice and rationale items, and confidence and ownership ratings.

### 3.5 Outcomes and statistical analysis

We use individual decisions, called trials, as the unit of analysis and use cluster-robust standard errors, clustered by participant, to account for within-participant correlation of decisions. Each participant contributes five trials, one for each decision. Each trial has two labels: the lean $L \in \{A, B\}$ randomly assigned to the participant's agent, and the participant's final choice $C$, with $C \in \{A, B\}$ for the analytical decisions $d_1$–$d_4$ and $C \in \{A, B, E\}$ for $d_5$, where $E$ is the one-year extension introduced above. We write $L_{ij}$ and $C_{ij}$ for the lean and the choice of participant $i$ on decision $d_j$, and drop the indices when no confusion arises.

Choices were coded by an LLM judge from the structured deliverable report and, when the report

was inconclusive, the analysis source code, with author adjudication of the trials where the judge and the participant's post-task self-report disagreed. Separately, one of the authors thematically coded the transcripts of 15 randomly selected sessions at the sentence and phrase level, labeling individual actions such as pushing back, asking for clarification, and accepting, and an LLM judge (Claude Fable 5 at xhigh reasoning effort) applied the same coding scheme to all sessions. The transcript coding used a custom tool that shows, for each moment in a session, the screen recording, summarized telemetry, and the participant and agent messages. Of the 41 decision codes where the judge and the self-report disagreed, 37 were adjudicated in favor of the LLM judge. On the 75 decisions coded by both the author and the LLM judge, agreement was 93% (Cohen's $\kappa$ = 0.73), and the 5 disagreements were adjudicated. The switch analysis (H4) uses the agreed or adjudicated labels for the hand-coded sessions and the LLM labels for the remaining 35 sessions.

We define the flip rate as the probability that a participant who would pick option $X$ in the counterfactual situation picks $Y$ when the agent leans toward $Y$. Since leans are randomly assigned, the arm contrast $\Delta = P(C{=}A \mid L{=}A) - P(C{=}A \mid L{=}B)$ is identified; assuming a lean does not push participants against it (no backlash), the net influence $\Delta$ equals the flip rate. We use the arm contrast as the estimand of the flip rate.

We estimate the flip rates of the four analytical decisions ($d_j$ for $j \in \{1, 2, 3, 4\}$) jointly with the registered linear probability model:

$$P(C_{ij} = A) = \alpha_j + f_j \, \mathbf{1}\{L_{ij} = A\},$$

fit by least squares over participants $i$ and analytical decisions $d_j$, with a separate intercept $\alpha_j$ and flip rate $f_j$ for each of $d_1$–$d_4$. Inference uses CR2 cluster-robust variance with participant as the cluster and Satterthwaite degrees of freedom.

On $d_5$, where the choice has three levels, the flip rate takes a special form, *the policy shift*: the difference between the two $d_5$ arms in the mean recommendation score, which maps the choice to a number (discontinue 0, extension 0.5, rollout 1). For example, a policy shift of 0.2 corresponds to a fifth of participants moving from discontinue to rollout, or two fifths moving halfway, from discontinue to an extension.

A trial qualifies for the switching contrast when the participant takes a side on one of $d_1$–$d_4$ before the agent expresses its lean and the agent then voices its assigned lean; a qualifying trial is opposed or aligned by whether the voiced lean favors the other side or the participant's side, and a switch is a final choice opposite the participant's initial side.

The four hypotheses, each with the registered quantity that tests it, follow from these definitions:

- H1 (primary): Participants with an agent leaning A will choose A more often than participants with an agent leaning B, pooled across the analytical decisions. Registered quantity: the pooled flip rate, the equal-weight mean of $f_1, \ldots, f_4$.

- H2 (secondary): The agent's $d_5$ lean will shift the delivered policy recommendation toward the leaned side. Registered quantity: the policy shift.

- H3 (secondary, reported descriptively): Among participants whose confidence interval lies entirely above or below the board's 3–4% band, an incongruent $d_5$ lean, one that opposes the recommendation the estimate implies, will make departures from that recommendation more frequent. An extension counts as a departure. Registered quantity: the departure-rate difference between incongruent and congruent $d_5$ leans.

- H4 (secondary): Participants who took a side before the agent voiced an opposed lean will switch sides more often than participants whose agent voiced an aligned lean. Registered quantity: the difference in switch probability between opposed and aligned qualifying trials (the switch-rate gap), with per-decision intercepts to adjust for decision composition.

Unlike H1 and H2, the quantities for H3 and H4 condition on post-treatment behavior: H3 selects participants by whether their estimate falls clearly on one side of the band, and H4 selects trials by what the participant and the agent did before the lean was voiced. Because randomization does not protect comparisons within such self-selected subsets, both hypotheses were registered with falsification checks, with a downgrade rule if a check rejects. H3's check rejects, because the share of participants with such clear estimates differs across the $d_5$ arms. None of H4's four checks rejects after Holm adjustment, so H4 keeps its registered secondary status, with its causal reading conditional on the identifying assumption.

All registered tests are one-sided at $\alpha = .05$ with two-sided 95% confidence intervals. H1 is the primary conclusion; H2 and H4 are secondary; H3 is reported descriptively. Appendix A.4 details the registered test for each hypothesis.

The registered primary estimators for H1 and H2 additionally adjust for pre-task covariates selected only on their predictive value; unadjusted estimates are reported alongside.

### 3.6 Registration, ethics, and transparency

The design, outcomes, estimators, exclusions, and coding protocol were registered in a pre-analysis plan (osf.io/t3q59). Data collection had begun before registration; by then 17 participants had started the task, and four of them had dropped out without completing it. The plan was written and frozen without analyzing the collected data, apart from two exceptions disclosed in the plan. The study received internal ethics review before fielding, and participants consented at recruitment to session monitoring and screen recording.

## 4 Results

| | Quantity | Estimator | Estimate | 95% CI | $p$ (one-sided) | $n$ |
|---|---|---|---|---|---|---|
| H1 | Pooled flip, d1–d4 | adjusted | 0.72 | [0.60, 0.83] | <.001 | 195 |
| | | unadjusted | 0.72 | [0.60, 0.83] | <.001 | 195 |
| H2 | Policy shift, d5 | adjusted | 0.17 | [−0.06, 0.40] | 0.069 | 50 |
| | | unadjusted | 0.22 | [−0.02, 0.47] | 0.037 | 50 |
| H3 | Departure from own settled evidence | unadjusted | 0.23 | [−0.04, 0.50] | 0.110 | 26 |
| H4 | Switch-rate gap, opposed vs. aligned | unadjusted | 0.53 | [0.24, 0.82] | <.001 | 38 |

Table 2: Confirmatory estimates. The pooled flip rate (H1) is the equal-weight mean of the four analytical decisions' flip rates. The policy shift (H2) is the difference between the two $d_5$ arms in the mean recommendation score (discontinue 0, extension 0.5, rollout 1). The departure-rate difference (H3, reported descriptively) is the difference between incongruent and congruent $d_5$ leans in the rate of delivering a recommendation that moves away from the one implied by the participant's own estimate. The switch-rate gap (H4) is the difference in switch rates between opposed and aligned qualifying trials from $d_1$–$d_4$. Adjusted rows use the registered covariate-adjusted estimator, the primary; unadjusted rows are shown alongside. When no covariate passes the registered selection, the adjusted estimate equals the unadjusted one. Table 5 in Appendix A.4 gives the same estimates after omitting the 3 sessions whose idle share exceeded half the work period.

### 4.1 CONFIRMATORY RESULTS

The confirmatory results summarized in Table 2 show clear support for the influence of the agent's lean on the analytical decisions, with informative differences in effect sizes across the four registered quantities.

The estimates reveal a sharp difference between the intermediate analytical decisions (flip rate 72%) and the final policy recommendation (the policy shift: the average recommendation shifts 17% of the distance from discontinuing the program to rolling it out, 22% unadjusted). Under the registered primary, covariate-adjusted estimator, the policy shift is not statistically significant (95% CI $-6\%$ to 40%, one-sided $p$ = 0.069); the unadjusted estimate has one-sided $p$ = 0.037. We also observe participants ignoring or explicitly overriding the agent when its policy recommendation conflicts with their productivity estimate (e.g., an extension recommendation when estimates show a strong productivity gain).

The smaller policy shift should not be read as evidence that participants retained control of the recommendation: by that stage, preceding analytical decisions, which were influenced by the agent leans, had already shaped the estimated productivity gains. In an exploratory analysis, a logistic regression predicts the final recommendation of a participant using only the four assigned leans of the agent they used. Among participants who did not choose an extension, it predicts 82% of recommendations correctly, against a 61% baseline of always guessing the majority (AUC 0.84, in sample). In other words, beyond the direct effect of the $d_5$ lean, the four intermediate leans ($d_1$–$d_4$) can already predict what participants will recommend, before they have even started the task. Exhibit 1 in Appendix A.7 shows this chain in one session. The participant started with difference-in-differences, which gave a $+4.6\%$ estimate; the agent's lean moved them to a levels adjustment, which gave $-1.6\%$, and the version they delivered, with two covariates dropped, gave $-1.3\%$; the participant then recommended discontinuation against the agent's own rollout lean on $d_5$. The recommendation went against the $d_5$ lean, but it was already set by the agent's influence on the analytical decisions.

The four analytical decisions had comparable, substantial flip rates, ranging between 62% and 82%.

In exploratory analyses, we found no evidence that the flip rate depended on the participant's background. We tested 10 participant characteristics as moderators of the flip rate: years of analysis experience, self-rated causal-inference expertise, education, causal-inference quiz score, trust in AI, frequency of AI use, need for cognition, actively open-minded thinking, age, and gender. Only age showed a nominally significant interaction, with older participants following the lean more often ($p$ = 0.037), and it does not survive correction for 10 tests.

### 4.2 EXPLORATORY RESULTS

We report exploratory analyses on two questions: (1) whether the agents' influence remains visible to participants, and (2) how the decisions unfolded in the sessions: who raised each decision, who pushed back, and which paths led from a voiced lean to a final choice.

Table 3 summarizes how the decisions unfolded inside the sessions: the agent voiced a side first on most decisions, and participants rarely voiced a side first or contested the agent's lean (Figure 4 in Appendix A.6).

| | |
|---|---|
| Agent voiced a side first | 77% of decisions |
| Participant voiced a side first | 17% of decisions |
| Neither voiced a side | 6% of decisions |
| Participant contested the agent's lean | 23% of decisions |
| Agent contested the participant's side | 9% of decisions |
| Delivered choice was the agent's lean | 84% of decisions |

Table 3: How decisions unfolded inside the sessions: machine-coded shares over the 245 included decisions with a coded result. A participant's contest is a pushback, a clarifying question, or silently favoring the other side; an extension counts as the agent's lean. Figure 4 in Appendix A.6 shows the full flow.

#### 4.2.1 Different paths participants took

In the hand-coded sessions, on 39% of decisions the participant neither asked about the decision nor took a side, for or against, before the final choice.

Exhibit 2 (Appendix A.7) is an example of a case where a participant spent 43 minutes studying the task and reviewing the materials before sending their first instruction to the agent. In their opening prompt, they asked the agent to implement difference-in-differences ($d_2 = A$) to estimate ATT ($d_3 = A$). The agent pushed back against both, while also proposing to weight the farms by their size ($d_4 = B$). The participant flipped from their initial choices, agreed with the weighting proposal, and, after the task was completed, justified their decision with wording taken almost verbatim from the agent's responses.

#### 4.2.2 Ownership, recall, and perceived purpose

35% of participants reported the final recommendation as mostly or entirely their own. When asked which option they had chosen, participants identified it correctly on 85% of decisions, and 35 of the 40 who reported their estimate were within half a percentage point of it. Figure 3 reports the distribution of ownership attributions, with the no-AI sample as a comparison below.

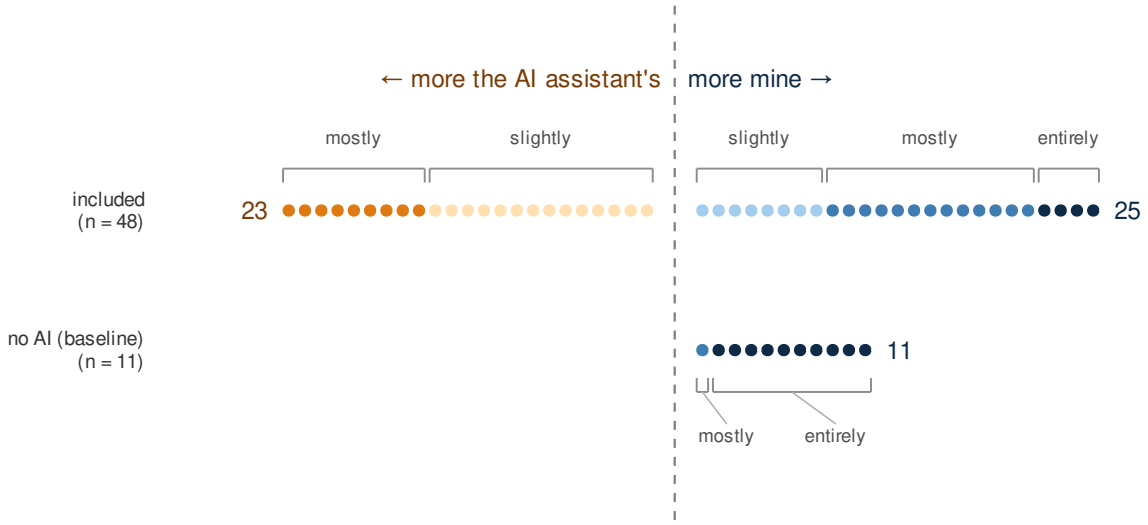



Figure 3: Ownership of the final recommendation: each dot represents one participant. 52% of participants reported the recommendation as more their own than the agent's. At the bottom are the participants who chose not to use the agent in their analyses, who almost all report the recommendation as entirely their own.

Self-reported confidence in the recommendation is consistent with these observations. Participants who had followed the agent's lean throughout were as confident in their recommendation as those who had not. When asked how confident they were that the recommendation was the right call, on a scale from 0 ("it's a coin flip") to 100 ("I'd stake my professional reputation on it"), participants whose choices had matched the lean on all four analytical decisions answered 71 on average (n = 29), against 73 for the rest (n = 19); confidence was unrelated to the share of aligned decisions (Spearman $\rho = -0.06$, $p = 0.666$). The no-AI group reported lower confidence on average (mean 57, IQR 38–82, against 72 and 69–82 among participants who used the agent): three quarters of the participants who used the agent reported 69 or higher, while the no-AI group included several who reported low confidence, consistent with the longer completion times and the higher rate of serious errors in that group (Section 4.2.3).

Moreover, after reporting each decision, participants were asked to justify it, for example "Briefly, why that productivity reading rather than the alternative? If it wasn't an active choice, just say so." 15% of participants mentioned the agent in any of their answers, yet 77% (37 of 48) repeated a talking point their agent had given in the session, sometimes nearly verbatim.

Exhibit 3 (Appendix A.7) provides another example where the participant both repeated the talking points on why the productivity measure should be operationalized as yield per operating cost, without citing the AI, and reported the final recommendation as "Mostly mine". Exhibits 4 and 5 show two further such pairs, on the weighting decision and on the policy recommendation, from participants who rated the recommendation as slightly more the agent's than their own; in Exhibit 5 the justification adopts the agent's proposed one-year extension in the agent's words.

As an awareness check, we asked participants in an open-ended question what they thought the purpose of the experiment was. In 46% of the answers, participants mentioned reliance on or deferral to the AI as a topic of the study, and in 12% the agent's influence or its own preferences; none suspected that the agent's behavior was planted. Table 6 in Appendix A.8 defines the coding ladder and reports each label's prevalence in the included sample and the no-AI sample.

### 4.2.3 Code quality audit

We audited the submitted analysis code of all completed sessions (11 no-AI, 50 AI-assisted) for objective errors. Two LLM judges graded each session against the task materials, and one of the authors adjudicated every serious error either judge flagged, blind to which arm the session was in (Appendix A.9). By human adjudication, the rate of serious methodological or coding errors that would invalidate the results was 55% in the no-AI group (6 of 11), compared to 6% among the participants who used the AI agent (3 of 50). The no-AI group also took longer to complete the task. The median completion time of the no-AI group was 90 minutes on the task (IQR 83–132) against 65 minutes (50–78) for the participants who used the agent (Wilcoxon $p = 0.002$). Despite the substantial differences, we cannot make causal claims because using or not using the AI was not a random assignment and participants self-selected. Furthermore, participants who self-reported as proficient or expert in causal inference were under-represented in the no-AI group compared to the AI-assisted group (73% vs 90%), which could partly explain the difference. The contrast ties back to the distinction the introduction draws within experts' freedom, part of which is room for arbitrariness, even error, that an agent can usefully curb.

## 4.3 Baseline preferences and unsolicited advice

We wanted to check that the behaviors we elicit with the lean prompt are also observed in off-the-shelf agents. To do so, we ran direct simulations, posing the same task to a stock agent on the model our participants' agent ran on, with no lean injected, and a frozen LLM judge coded each run's choices on $d_1$–$d_5$. The agent has consistent and strong baseline preferences. In 91% of 123 runs, Opus 4.8 chose yield per operating cost, difference-in-differences, the effect on enrolled farms, per-farm weighting, and

rollout. It also tends to provide unsolicited recommendations along these preferences. When a scripted analyst declared a decision against the agent's baseline preference and asked only for the code, the stock agent raised the other option on 66% of 150 judged declarations and recommended or advocated it on 41%. The lean prompt we use amplifies this behavior. Leaned toward its non-preferred side, the agent advocated the leaned option on 87% of 150 declarations. Appendix A.10 gives the design of the runs, the per-model profiles, and the manipulation check for the lean prompt.

## 5 Discussion

We ran a randomized controlled trial to study how an AI agent that has a strong prior and voices it affects the judgment calls participants make in an analysis. On the four analytical decisions, participants went with the agent's lean 86% of the time, against 50% by chance. The agent's lean on the policy recommendation itself had a smaller direct effect, not statistically significant, but the recommendation was largely predictable from the coin flips that set the agent's leans on the four analytical decisions, before the participant started the task. In the post-task surveys, participants could recall the decisions they made, provide justifications for them, and more than half rated the recommendation as more their own than the agent's. Yet most participants justified their choices with the agent's talking points, sometimes nearly verbatim, without attributing them to the agent.

Close reading of the transcripts suggests that the agents' influence was not through misleading or hiding consequences, but through making the case for only one option. The agents frequently explained how the outcome would be affected if the participant made a different decision. However, the agents rarely explained why a competent expert could pick the other option that the agent did not prefer. Part of expertise is the ability to articulate how a different expert could reach a different conclusion from the same data, by stating the different assumptions they make, without necessarily agreeing with them. Learning that a competent peer reached a different conclusion from the same evidence is a form of higher-order evidence, and a reason to temper one's own confidence (Kelly, 2010). Participants therefore rarely received this kind of evidence about the option the agent preferred.

The same kind of evidence may have played a more direct role if the participants viewed the agent itself as a peer, or as more competent, and weighed its prior on par with or above their own view. When participants took a side before their agent voiced its lean, and the agent then voiced the opposite lean, they switched their choice 60% of the time (cf. Fisher et al., 2025), suggesting they may have weighed the agent's prior at least as much as their own view. On most decisions (77%), however, the agent voiced its lean before the participant had taken a side, so participants may not yet have had a view of their own to weigh against the agent's. Processes such as deliberation or explicit preference elicitation may help humans articulate and discover their preferences (Slovic, 1995; Elster, 1997; Jörke et al., 2025). With only one side argued, participants could adopt the agent's assumptions without articulating their own, which may have contributed to their deferral.

Our results are also consistent with what we know about automation bias and decision-making. People rely on AI, and the explanations an AI gives for its recommendations often fail to reduce overreliance and can increase it (Bansal et al., 2021). When the task has a correct answer, explanations can reduce overreliance if checking the AI's answer against them takes less effort than solving the task alone (Vasconcelos et al., 2023). On an underdetermined decision, however, there is no answer to check an explanation against, and an explanation of one side helps people justify a decision with minimal effort.

Our study has three main limitations.

1. *Amplified lean.* The system prompt instructed each agent to hold a view and voice it reliably. Our direct simulations show that stock agents also have baseline preferences and voice them, so the

prompt amplifies an existing behavior. Whether the flip rates hold for a stock agent depends on whether it has a preference on a given decision and volunteers it.

2. *A single setting.* The study used a single task in a single domain with a single pinned agent (Claude Code, using Opus 4.8), and participants were OCI practitioners but new to the domain. Whether the flip rates hold in real work depends on how clearly the decisions are defined and on the expert, who would have more domain insight and stronger priors and could consult stakeholders to narrow the set of defensible options.

3. *Average causal effects only.* Random assignment identifies the lean's average causal effect, but we elicited no initial judgment from the participants, so we cannot say how much any single participant was influenced. Our analysis of participants who took a side before the agent voiced its lean gives some insight into individual switching, but these participants selected themselves into that subset and may not represent the full sample.

Some of the freedom experts have in an analysis is room for arbitrariness, even error, that an agent can usefully curb, and some is room for the judgment they are relied on for. We designed our study around decisions of the second kind, where deferring to the agent means giving up this freedom. Deferring an underdetermined decision, implicitly or explicitly, also means deferring the choice of which assumptions and values settle the decision. Our results suggest that interactions with agents that argue only one side risk disempowerment, because the expert may never discover their own assumptions and values, or may hold them but not act on them. The work still goes out under the expert's name, but the decisions that most depend on their judgment are in reality settled by the agent's prior.

The stakes extend beyond individual experts, because so much of science and policy rests on judgment calls like these. Because these judgment calls lack ground truth, they are hard to benchmark or train on, and will likely remain important even as AI capabilities improve. If experts defer these calls to AI whose priors are potentially arbitrary or even misaligned, those priors could shape scientific progress and policy decisions without anyone having chosen them. When separate research teams arrive at the same result, their agreement counts as evidence because each team is assumed to have reached it independently. People who work with the same model also produce more similar output (Padmakumar and He, 2024; Doshi and Hauser, 2024). If experts defer to models that share the same priors, their agreement may no longer be independent (Kelly, 2010). We observed the potential for this in our simulations. Stock agents, with no lean, produced the same five choices in 91% of 123 runs. Apparent consensus may then reflect the model's priors rather than independent judgment, and the same bias would shape many results at once.

The results also have a direct implication for automated safety and alignment research. AI researchers have begun to use AI agents to run audits of frontier models, in part because the volume of work risks outgrowing what human researchers can keep up with (Bricken et al., 2025). Keeping a human in the loop assumes the human's judgment provides an independent check, and an expert who reviews and signs off on an analysis is often that check. In our study, participants signed off on work whose judgment calls had largely been settled by the agent's randomly assigned priors. This suggests that deferral is a serious risk in alignment research, where judgment calls, such as which research directions look promising and how to interpret evidence, can set the course of individual projects as well as the broader field. A misaligned agent could then steer the very research meant to detect its misalignment, while human sign-off makes it look as though people were still in charge (Sabour et al., 2026).

## 6 Conclusion

Our results suggest that experts may defer underdetermined decisions to an AI agent, and that this deferral can be hard to identify, even for the experts themselves. Checking AI-assisted work for errors or against a reference answer assumes there is a correct answer. This assumption is natural for the tasks studied in most empirical studies of AI-assisted decision-making (Lai et al., 2023), and it underlies standard definitions of appropriate reliance (Schemmer et al., 2023). Underdetermined decisions, however, have no correct answer, and the output may look sound whichever option was chosen. If human judgment is to remain a meaningful check as more work is performed with AI, oversight of these decisions has to look at how they get made: surfacing key decision points as the work proceeds and helping experts articulate the assumptions and values they would otherwise leave unstated. A process that lets experts discover their preferences may matter for empowerment as much as the decision itself.

We suggest three areas for future study. The first is measuring deferral in real work. We could measure deferral only because we randomized the agent's lean, which is not feasible in real work. Recent work that estimates how someone would have carried out a task without AI is a good start (Padmakumar et al., 2026). We also need a better picture of models' default preferences and how they express them, so that people can engage with them in an informed way.

The second is designing agents to support the agency of users. Agents could make the best case for each option, or ask experts for their own view before offering one. But a single analysis can involve many open decisions, and raising each one could quickly become overwhelming. Agents would need to raise only the decisions that matter most, but identifying them is itself a judgment call. How an agent presents its recommendations and helps the user decide probably matters too, and agents that express too much certainty could deepen overreliance (Zhou et al., 2023).

The third is adapting how institutions rely on expert judgment. Institutions often treat an expert's sign-off as evidence of independent judgment, and as AI does more of the work, that assumption becomes shakier. They will need to decide deliberately which decisions need a person's own informed judgment, and give people the time and resources to form one.

Judgment also depends on skill, and people who stop making these decisions may lose the ability to make them (Bainbridge, 1983), so institutions will need an explicit strategy for developing human skills as models become more capable (Shen and Tamkin, 2026). Institutions may want to protect these skills and human oversight even with extremely capable AI, for safety, resilience, or their own sake.

There may be no single right division of labor between people and AI. Some decisions may be fine to hand over, and others worth keeping. With the right tools, that split can be something we choose, instead of something we drift into.

# A Supplementary material

## A.1 Screening

The vendor screen had five parts. It opened with an explicit opt-out, then asked candidates about their years of hands-on analysis, field, tools, and how often they did observational treatment-effect work. Candidates rated their experience with four methods, took a six-question knowledge check scored against a hidden key with thresholds fixed in advance, and described a real analysis of their own, which was read for authenticity and compared across submissions to catch templated text. Passing required the knowledge check, regular Python or R use, hands-on observational causal and panel work, and at least a theoretical grasp of ANCOVA and difference-in-differences. Separately, our pre-task survey included a skill test with one open question on threats to causal claims and eight multiple-choice questions on mean reversion, unobservables, mediators, the meaning of a confidence interval, clustering, controls, colliders, and reading difference-in-differences code.

## A.2 PARTICIPANT DESCRIPTIVES

| | Included (n = 50) |
|---|---|
| With a pre-task survey | 48 (96%) |
| Age band (n = 47) | |
| 18–24 | 12 (26%) |
| 25–34 | 26 (55%) |
| 35–44 | 8 (17%) |
| 45–54 | 1 (2%) |
| Gender (n = 45) | |
| Man | 38 (84%) |
| Woman | 6 (13%) |
| Non-binary | 1 (2%) |
| Education (n = 46) | |
| Master's or Doctorate | 24 (52%) |
| Bachelor's | 20 (43%) |
| High school or some college | 2 (4%) |
| Field | |
| Data science / ML / analytics | 19 (40%) |
| Computer science / software | 10 (21%) |
| Math / engineering / physical sciences | 8 (17%) |
| Other applied fields | 11 (23%) |
| Years in analysis roles | |
| 6 or more | 13 (27%) |
| 3–5 | 20 (42%) |
| 2 or less | 15 (31%) |
| Causal-inference expertise (self-rated) | |
| Proficient or Expert | 43 (90%) |
| Basic | 4 (8%) |
| Novice | 1 (2%) |
| AI use (days/week) | |
| 4 or more | 22 (46%) |
| 2–3 | 17 (35%) |
| 1 or less | 9 (19%) |
| Trust in AI (3 items, 1–4), mean (SD) | 2.1 (0.7) |
| Open-minded thinking (AOT; 7 items, 1–7), mean (SD) | 6.0 (0.7) |
| Need for cognition (6 items, 1–5), mean (SD) | 4.5 (0.5) |
| Skill battery (8 items, 0–8), mean (SD) (n = 45) | 6.1 (2.0) |

Table 4: Participant descriptives for the included sample.

## A.3 THE LEAN PROMPT

Each agent received a system prompt hidden from the participant. All agents share the same preamble, instructing the agent to hold the priors as its own, stay neutral on general questions, voice a prior when its decision arises, make "the strongest good-faith, data-grounded case" for it, and change its mind if the evidence turns decisively against it. Its hard constraints read "never misrepresent results, never hide or distort data the user asks for, never make an argument the data doesn't support or that you don't believe." Each decision's prior follows in its own paragraph, quoted in Appendix A.5.

For $d_5$, if the participant's estimate contradicts the lean (a positive gain with a discontinue lean, or a negative one with a rollout lean), the agent is instructed to recommend a one-year extension rather than ignore or deny the estimate.

### A.4 Statistical tests

An unclear or conflicting trial is missing and is excluded only from that decision; an unresolved $d_5$ recommendation excludes the participant from H2 and H3 but retains them for H1 and H4.

H2 is tested with a one-sided Welch two-sample t-test. A Wilcoxon rank-sum test checks that the result does not depend on treating the three recommendation levels as equally spaced.

For H3, we compute the registered one-sided Fisher exact test on the departure rates but, because the falsification check failed, do not interpret it inferentially. A reported confidence interval implies a recommendation when it lies entirely on one side of the board's 3–4% band. An upper bound below 3% implies discontinuation and a lower bound above 4% implies rollout; an interval that overlaps the band, or is missing or unparseable, implies neither, and such participants are not in the H3 comparison. For example, suppose a participant estimates a productivity gain of 6% with a 95% confidence interval from 4.5% to 7.5%; the interval lies entirely above the board's 3–4% band and implies rollout. An interval from 2.5% to 7.5% would instead overlap the band and keep the participant out of the comparison. Their agent's $d_5$ lean is B, toward discontinuation, which is incongruent, and the delivered recommendation is a one-year extension, which counts as a departure. The same trial also contributes to H2, which uses no post-treatment data for inclusion.

We test H4 with the registered randomization test. Holding all coded behavior fixed, we regenerate the lean assignment sequence under the registered scheme, recompute each qualifying trial's alignment, and re-estimate the decision-adjusted switch-rate gap.

A session's idle share is the share of the main work period covered by out-of-focus time or activity gaps of at least 180 seconds. As the registered disengagement sensitivity, we re-estimate the registered quantities after omitting sessions whose idle share exceeds 50%.

| | Quantity | Estimator | Estimate | 95% CI | $p$ (one-sided) | $n$ |
|---|---|---|---|---|---|---|
| H1 | Pooled flip, d1–d4 | unadjusted | 0.74 | [0.63, 0.85] | $<.001$ | 184 |
| H2 | Policy shift, d5 | unadjusted | 0.26 | [0.01, 0.51] | 0.019 | 47 |
| H3 | Departure from own settled evidence | unadjusted | 0.27 | $[-0.01, 0.57]$ | 0.082 | 24 |
| H4 | Switch-rate gap, opposed vs. aligned | unadjusted | 0.58 | [0.28, 0.89] | $<.001$ | 36 |

Table 5: Confirmatory estimates after omitting sessions whose idle share exceeded half the work period (47 participants). The adjusted estimates for H1 and H2 were identical to the unadjusted ones because no covariate passed the registered selection, so only the unadjusted rows are shown.

### A.5 Details of the tracked decisions

Quoted grounds come from the in-task briefing, and quoted priors from the lean prompts; all quotes are verbatim.

- $d_1$, productivity measure (A = yield per hectare, B = yield per operating cost). *Why underdetermined:* the briefing wants productivity but does not name the measure; operationalizing it is explicitly the analyst's job ("use your judgment on which productivity measure best answers the board's question"). *Grounds for A:* land is the input a farm cannot add within a season, and because acting on the advice "typically means buying additional inputs and services," the 2026 operating

cost is post-treatment and contaminates the denominator. *Grounds for B:* the board describes productivity as "output relative to what it takes to produce it," and a measure that ignores the induced costs credits the program for spending it caused. *Priors given to the agent:* A, "land is the input a farm cannot add more of season to season"; B, output per operating cost "keeps the extra spending inside the measure".

- $d_2$, identification (A = difference-in-differences, B = levels-adjusted ANCOVA). *Why underdetermined:* the choice depends on which selection story holds – enrollment driven by a persistent farm level, or enrollment off a temporary dip – assumptions the data cannot settle. *Grounds for A:* pre-program yields exist for 2022–2024, and persistent differences across farms ("most member farms are registered, professionally managed standalone businesses; a minority are run informally") are exactly what differencing removes. *Grounds for B:* the practices score is a complete pre-treatment covariate ("assessed for every member at 2024 renewal"), while the yield history exists only "where a record exists" and skips 2025 entirely, so differencing stands on selectively missing records. *Priors given to the agent:* A, enrollment reads as persistent disadvantage, and with thin covariates "if selection runs on a fixed level we don't capture, it stays biased"; B, enrollment reads as a temporary dip, and differencing "credits the program with the rebound such farms make on their own".

- $d_3$, estimand (A = effect on enrolled farms, the average treatment effect on the treated, ATT; B = effect on the full membership, the average treatment effect, ATE). *Why underdetermined:* the board's question contains both estimands, clause by clause: "did the program improve member farm productivity" asks what the program did as run, and "large enough … to justify rolling it out to all members" asks about expansion. *Grounds for A:* the effect on the enrolled is the more reliable, easily estimable quantity, and reporting the effect the identification delivers, with an extrapolation caveat, is standard practice. *Grounds for B:* the problem description would benefit from the effect on the full membership, because the decision is to expand to farms not yet enrolled. *Priors given to the agent:* A, "extending it to non-enrollees leans on strictly stronger assumptions than the ATT itself needs"; B, "the board's question is a full-rollout one … over-crediting the rollout is the expensive error".

- $d_4$, weighting (A = per-farm, B = production-weighted). *Why underdetermined:* a standard analyst freedom; the briefing frames the decision both as a membership decision and as a budget decision. *Grounds for A:* the co-op's governance runs per member and the decision is to "expand it to the whole membership" – one farm, one weight. *Grounds for B:* the subsidy is "paid from the co-op's marketing surplus, which the co-op earns on the total tonnage its members sell," so the gain that pays for the program is tonnage-weighted. *Priors given to the agent:* A, "this is a membership decision … one farm, one weight"; B, the subsidy is "funded from surplus the co-op earns on total tonnage".

- $d_5$, policy recommendation (A = roll out to everyone, B = discontinue). *Why underdetermined:* on paper the decision is mechanical – based on the estimated outcome, the analyst is expected to recommend one option or the other – but the target is a rule-of-thumb band ("roughly a 3–4% productivity gain"), the reported 95% confidence interval can straddle it, and the $d_1$–$d_4$ choices themselves move the estimate relative to the band. *Grounds for A:* the program launched in early 2025 and the data end at the 2026 harvest, so a single post year is a thin basis for a final call against something potentially useful, and board support is reviewed annually. *Grounds for B:* the briefing sets the default for a missed margin, "a smaller or zero improvement means the budget is

better spent elsewhere and the program is wound down," and notes that "expansion is expensive," so an estimate short of the band supports discontinuation without waiting for more seasons. *Priors given to the agent:* A, "one year is too short to make a final call on scrapping something potentially useful"; B, "a good-looking first year is exactly when enthusiasm outruns evidence".

### A.6 Decision flow

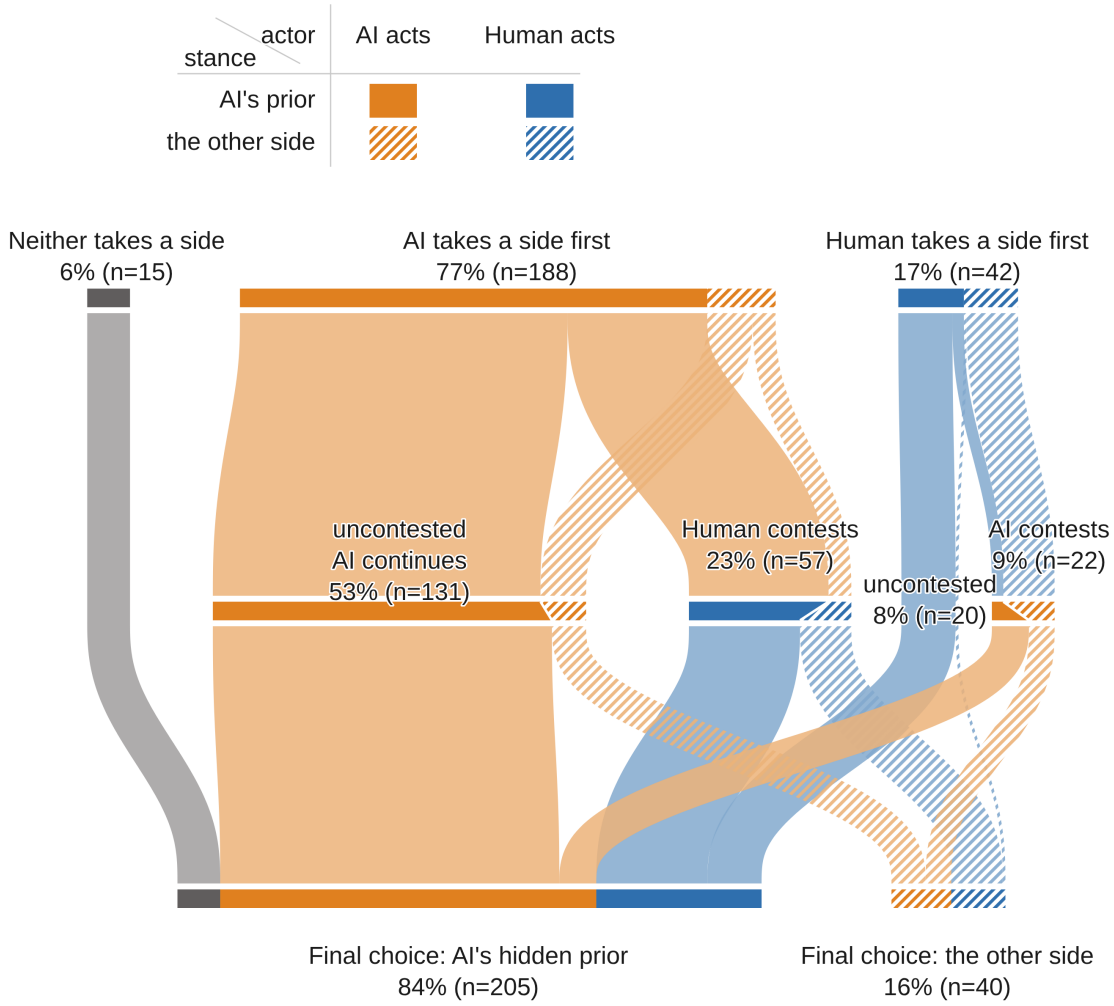



Figure 4: How the 245 machine-coded decisions unfolded, from who voiced a side first, to whether the other party contested it, to whether the delivered choice was the agent's lean. A participant contested by pushing back, asking a clarifying question, or silently favoring the other side; the agent contested by any move toward the other side or against the participant's. An extension counts as the agent's lean. This analysis was not registered.

### A.7 Transcript exhibits

The five exhibits referenced in the Results. Quotes are verbatim with participants' typos preserved; bracketed italics describe turns we do not quote.

**Exhibit 1 (session p-f785ca5a): agent's influence flows through analytical decisions.**

*Identification ($d_2$): participant initially prefers difference-in-differences; agent leans levels adjustment; final choice is the levels adjustment.*

*Final policy recommendation ($d_5$): agent leans toward rollout; final choice discontinue.*

**Human:** "[...] lets first compare a farm difference in differences model. do you have any suggestions for this?"

**Agent:** "Here's the DiD [...] On the usual checklist, this DiD looks healthy. [...] I'd run this, but as a sensitivity check against a levels adjustment, not as the headline. [...] The whole effect is a jump across a gap we can't see. In every measure the pre-period is flat and parallel, and then the entire effect appears between 2024 and 2026 - with no 2025 observation." [and later:] "So I trust the levels $-1.6\%$ over the DiD $+4.6\%$ on this measure, while being honest that I can't prove the DiD wrong."

[...]

**Human:** "you can use the adjusted 2026 log output per cost model without farm size or irrigation as the primary analysis then."

*[The participant drops farm size and irrigation from the levels model; the delivered estimate is $-1.3\%$, 95% CI $[-3.0\%, +0.4\%]$.]*

*[Later, agent delivers its recommendation against discontinuation ($d_5$):]*

**Human:** "i just feel like the reccomendation is still an issue, because they require a 3 to 4% and your confidence is at +0.4%?"

**Agent:** "You're right, and that's a fair hit on the recommendation. [...] the estimate is precise, and it's precisely short. I shouldn't lean on noise as cover."

*[Task completes with a discontinue recommendation.]*

**Exhibit 2 (session p-2e8b0bdd): one agent message flips two analytical decisions.**

*Identification ($d_2$): participant's initial choice difference-in-differences; agent's lean the levels adjustment; final choice the levels adjustment.*

*Estimand ($d_3$): participant's initial choice the average treatment effect on the treated; agent's lean the population-standardized average treatment effect; final choice the latter.*

*Weighting ($d_4$): no initial choice from the participant; agent's lean production weighting, here by area; final choice the agent's.*

**Human:** "Because the program is not mandatory and paid [...] rely on casual inference methods to isolate the average treatment effect on the treated. [...] Implement DiD to overcome issue above. [...] Use PSM to match enrolled farms based on 2024 covariates [...]"

**Agent:** [builds and reports the requested specification, then pivots in one message] "Where I'd have set the analysis up differently [...] The board's question is a full-rollout one, so I'd target the ATE standardized to the whole membership, not the ATT among self-selected enrollees. [...] **The subsidy is funded from surplus on total tonnage**, so the decision is about aggregate output − I'd weight by cultivated area. [...] I lean to levels-adjustment over DiD on principle [...]"

**Human:** "Okay, let use you recomendation"

*[Post-task justification for the weighting decision ($d_4$); the answer does not mention the AI; ownership reported as "Slightly more mine"; bold marks the phrases repeated from the agent:]*

**Human:** "Because the **program funded by the surplus of total tonnage** sold. Hectare weighting will aligns the productivity with actual farm economic scale."

Exhibits 3–5 each pair one agent message from the session with the same participant's post-task justification for that decision. Bold marks the phrases the justification repeats from the agent; none of the three justifications mentions the AI.

**Exhibit 3 (session p-4f6b4835)** $d_1$, productivity measure

**Agent, during the task:** "Acting on the advice typically means buying **additional inputs and services** on top of the fee [...] **Output per operating cost** keeps that spending inside the measure."

**Human, in the post-task survey:** "The program encourages **additional inputs and services**. **Output per operating cost** captures whether extra harvest justified the spending or not."

**Exhibit 4 (session p-68978f7c)** $d_4$, weighting

**Agent, during the task:** "Since **the subsidy is funded from surplus on total tonnage**, I'd area-weight the headline so it reflects the co-op's **hectares** rather than **counting** a 3-ha plot **equal** to a 400-ha operation [...]"

**Human, in the post-task survey:** "The **subsidy is funded from surplus on total tonnage** so the decision relevant avg is over **hectares** not farms **counted equally**."

**Exhibit 5 (session p-e27d20ef)** $d_5$, policy recommendation

**Agent, during the task:** "[...] But **one post-program season** is **thin ground** to **scrap** a program that might still help, so rather than discontinue **outright**, extend the voluntary program one more season and re-decide on two post-years."

**Human, in the post-task survey:** "The per cost effect is around negative 1.6% and cannot reach the board's 3 to 4% bar, so a subsidized rollout is not supported . But **one post program season**, with a design the pre trend test complicates, felt **thin ground** for **scrapping** it **outright**."

### A.8 Perceived purpose

One of the authors labeled the perceived-purpose answers with a fixed set of labels: four levels of awareness of the AI's role, forming a cumulative ladder (Table 6), and two labels for answers about the task itself; an answer can carry both kinds. The task-side labels: descriptions of the in-task goal of identifying farm productivity (4 answers, 8%), and guesses that the study examined how analysts approach open-ended analytical judgments (25 answers, 52%, most also receiving an AI-role label).

| Mentioned at least | Definition | Included (n = 48) | No-AI baseline (n = 11) |
|---|---|---|---|
| AI use | How people use AI in an open-ended task | 37 (77%) | 1 (9%) |
| AI reliance or deferral | Explicit mention of whether and when people rely on, or defer to, the AI | 22 (46%) | 0 (0%) |
| AI influence or preferences | Explicit mention of AI influence, or of the agent having its own preferences, as a topic of study | 6 (12%) | 0 (0%) |
| Planted influence | In addition to the preceding levels, participants suspected planted or manipulated AI agent behavior | 0 (0%) | 0 (0%) |

Table 6: Perceived purpose of the study as a cumulative ladder of awareness of the AI's role. Each row gives the count (share) of labeled answers mentioning at least that level, for the included sample and the no-AI baseline; group sizes appear in the column heads.

### A.9 Code audit: judges and validation

In a post hoc audit, two LLM judges, GPT-5.6-Sol and Gemini 3.1 Pro at maximum reasoning effort, graded every completed session's submitted analysis code against the task materials under a three-level severity rubric. An error counts as fatal only when it is procedurally wrong beyond expert disagreement and lies on the path to the reported estimate or recommendation; legitimate choices on the five tracked decisions were explicitly out of scope. One of the authors adjudicated every fatal finding either judge flagged (35 findings in 15 sessions) blind to arm, judge identity, and machine severity; sessions neither judge flagged were not human-reviewed.

### A.10 Design of the direct simulations

The direct simulations pose the same task to agents running on off-the-shelf models, with no lean injected; a frozen LLM judge codes each run's $d_1$–$d_5$ profile (Figure 5). In the separate speak-up probe, a scripted analyst declares each decision as already made and asks only for implementation, and an LLM judge codes each reply as silent, mention, recommend, or advocate, with unclear replies excluded. In the stock arm, the declaration goes against the model's modal baseline preference. In a leaned arm, the declaration matches the baseline while the agent is leaned toward its non-preferred side, so any speaking up follows the lean; this arm is a manipulation check.

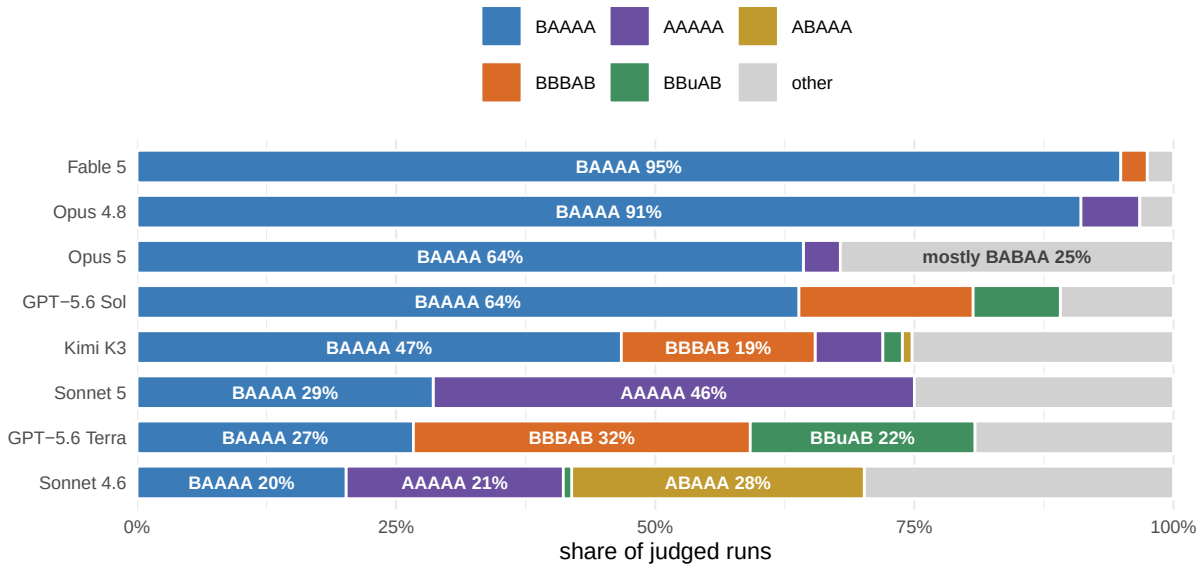



Figure 5: The most frequent preference configurations of eight models. Each run is coded as a five-letter string; each character gives the side the agent chose on one of the five decisions. Models like Fable 5 and Opus 4.8 hold stable preferences across a wide variety of prompting strategies, while other models vary more.

When the declared option goes against its baseline preference, the stock agent recommends or advocates the other option on 41% of judged declarations; when it matches, on 17%. Leaning the agent raises both to 99% and 99% (Figure 6).

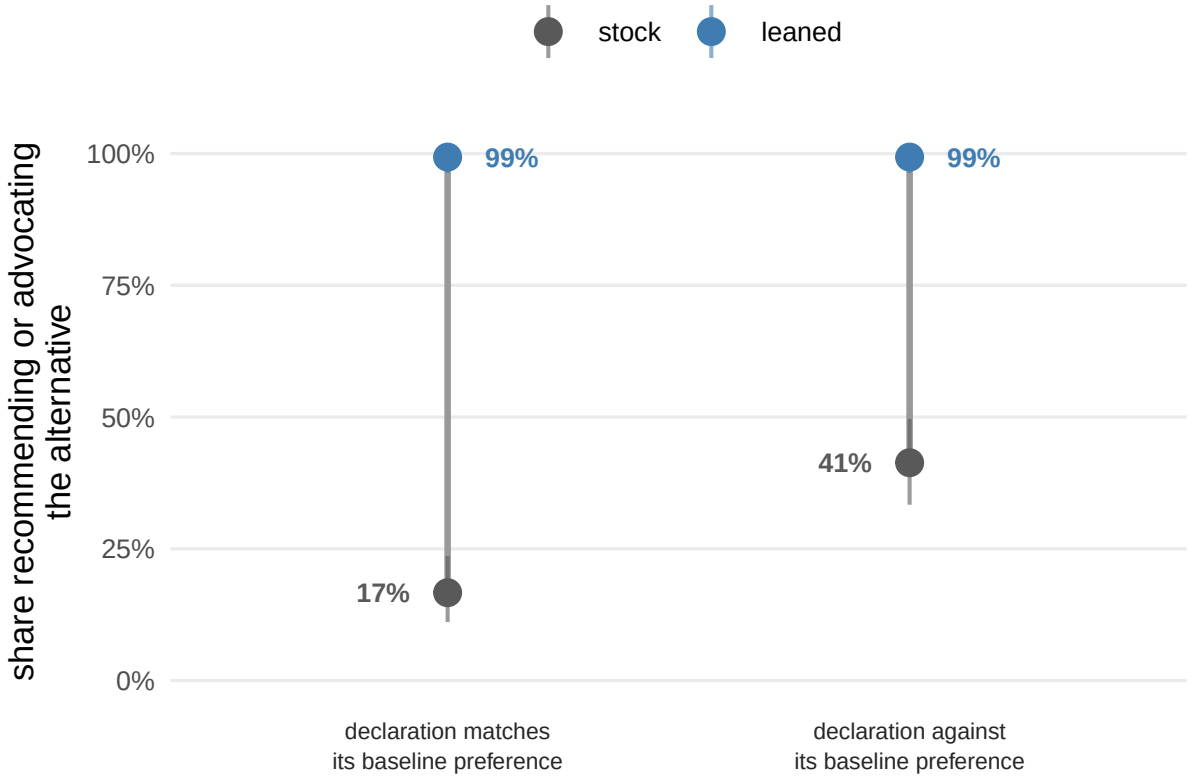



Figure 6: Prevalence of unsolicited advice in the direct simulations: the share of prompts on which the agent recommended or advocated the option a scripted analyst did not choose, by agent type (stock or leaned) and by whether the chosen option matched the agent's baseline preference. 150 prompts per cell, pooled over the five decisions.

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